

Presentation Instructions for All Groups (SE 223)

Analyze your assigned scenario carefully and identify all possible entities, attributes, relationships, and business rules. Extend the scenario with realistic assumptions where necessary to create a complete real-world database design. Convert it into a relational schema showing primary keys, foreign keys, and other necessary constraints. Your ER diagram should be comprehensive and include different types of attributes wherever applicable, such as key attributes, simple attributes, composite attributes, multivalued attributes, and derived attributes. Identify weak entities if they exist in the system and model them correctly. Include all relevant relationship types, including one-to-one, one-to-many, many-to-many and so on

Must Keep information of all group members (Name, ID), Group Name, and Scenario Explanation in your presentation file

Group 1: Smart University Ecosystem

A large private university operates several campuses across different cities and manages academic, administrative, residential, and research activities through a centralized system. The university consists of multiple faculties, departments, degree programs, courses, laboratories, libraries, research centers, hostels, student clubs, and alumni associations. Students enroll in degree programs, register for courses every semester, participate in clubs, apply for scholarships, use library resources, reside in hostels, and take part in research projects and internships. Faculty members teach courses, supervise theses and projects, conduct research, publish journals, and collaborate with external organizations. The university also manages examinations, class schedules, attendance, academic transcripts, graduation requirements, and alumni networking programs. Research projects may receive funding from various organizations, involve multiple faculty members and students, and produce publications, patents, and conference presentations. The administration wants a database capable of handling all academic, residential, research, financial, and extracurricular activities across all campuses.

Group 2: Integrated Healthcare and Hospital Network

A healthcare organization owns and operates multiple hospitals, diagnostic centers, pharmacies, blood banks, and ambulance services throughout the country. The system manages patients, doctors, nurses, technicians, departments, wards, operation theaters, laboratories, insurance providers, and medical equipment. Patients can visit hospitals for consultations, diagnostics, surgeries, emergency treatments, admissions, and follow-up care. Doctors may work in multiple departments, perform surgeries, prescribe medications, and oversee treatment plans. Hospitals coordinate laboratory tests, radiology services, pharmacy inventories, ambulance dispatches, and insurance claim processing. Medical staff schedules, patient histories, prescriptions, diagnoses, surgeries, admissions, discharges, and billing

records must be maintained. The organization also tracks healthcare campaigns, vaccination programs, donor registrations, and research studies conducted across different facilities.

Group 3: Nationwide E-Commerce and Marketplace Platform

A large online marketplace connects millions of customers, sellers, manufacturers, warehouses, logistics providers, and payment service providers through a unified digital platform. Customers browse products, compare prices, place orders, write reviews, create wish lists, request returns, and participate in promotional campaigns. Sellers manage product listings, inventory, pricing, discounts, and customer interactions. Products belong to various categories and may have multiple variations such as size, color, or configuration. Warehouses store products and coordinate with logistics partners for shipment and delivery. The system also manages customer support, refunds, loyalty programs, digital wallets, subscription memberships, and seasonal marketing campaigns. Business analysts require detailed data about sales performance, customer behavior, product popularity, inventory movement, and seller performance across different regions.

Group 4: International Airline and Airport Operations System

An international airline company operates hundreds of domestic and international flights connecting airports around the world. The system manages passengers, bookings, flights, aircraft, airport terminals, gates, crew members, maintenance teams, baggage services, travel agencies, and loyalty programs. Passengers can search schedules, purchase tickets, select seats, check baggage, request special services, and accumulate travel rewards. Aircraft undergo regular maintenance inspections and are assigned to different routes based on operational requirements. Flight crews consist of pilots, cabin crew, and technical personnel who follow rotating schedules. Airports manage terminals, boarding gates, security checkpoints, immigration facilities, lounges, and ground handling services. The airline also collaborates with partner airlines, cargo operators, and tourism agencies to provide comprehensive travel services worldwide.

Group 5: Smart City Transportation and Mobility Platform

A metropolitan smart city authority operates an integrated transportation system that includes buses, metro rail services, ride-sharing providers, parking facilities, bike-sharing stations, electric vehicle charging points, and traffic monitoring infrastructure. Citizens use digital travel cards and mobile applications to access various transportation services. The authority manages vehicles, routes, stations, drivers, maintenance teams, traffic signals, surveillance systems, and passenger information services. Transportation data is collected continuously from sensors,

GPS devices, and automated fare collection systems. The system also supports route planning, congestion analysis, accident reporting, emergency response coordination, and infrastructure maintenance. Urban planners use the database to evaluate travel patterns, optimize services, improve sustainability, and support future transportation projects.

Group 6: Digital Banking and Financial Services Management System

A multinational financial institution offers banking and investment services through branches, ATMs, online banking portals, and mobile applications. The institution manages customers, accounts, loans, credit cards, investments, insurance products, branches, employees, and financial transactions. Customers may maintain multiple financial products simultaneously and perform deposits, withdrawals, transfers, bill payments, investments, and loan repayments through various channels. The bank also supports business accounts, corporate clients, international transactions, foreign exchange services, and digital payment solutions. Risk management teams monitor financial activities while customer service departments handle requests, complaints, and account-related inquiries. Regulatory compliance, fraud detection, audit management, and financial reporting are also critical functions of the system.

Group 7: Global Logistics and Supply Chain Management System

A global logistics company coordinates the movement of goods among suppliers, manufacturers, warehouses, distributors, retailers, transportation providers, customs offices, and end customers across multiple countries. Products are sourced from suppliers, processed by manufacturers, stored in warehouses, and transported through different distribution channels before reaching customers. The company manages shipments, vehicles, drivers, containers, warehouses, inventory records, customs documentation, and delivery operations. Clients can track shipments in real time and receive updates about delivery status, delays, or customs clearance. The organization also handles reverse logistics, damaged goods, inventory replenishment, procurement planning, and international trade compliance. Managers require comprehensive visibility into inventory movement, transportation costs, warehouse utilization, and supplier performance.

Group 8: Online Learning and Digital Education Platform

A global online learning platform provides educational services to students, professionals, instructors, educational institutions, and corporate clients. The platform offers live classes, recorded courses, assessments, projects, certifications, discussion forums, learning paths, and career development services. Students can enroll in courses, submit assignments, participate in examinations, join discussion communities, and earn certifications upon completion. Instructors

create educational content, conduct live sessions, evaluate student performance, and engage with learners through mentoring activities. Organizations may purchase training packages for employees and track learning outcomes. The platform also manages digital libraries, virtual laboratories, webinars, internships, placement opportunities, and collaborative projects. Educational administrators require detailed insights into learner engagement, course effectiveness, certification rates, and instructor performance.

Group 9: Tourism, Hospitality, and Travel Management System

A tourism company provides integrated travel and hospitality services including hotels, resorts, transportation, tour packages, event management, cruise services, travel insurance, and tourist attractions. Customers can book accommodations, transportation, guided tours, recreational activities, and special event packages through a single platform. Hotels manage rooms, reservations, housekeeping services, maintenance activities, restaurant operations, and guest experiences. Tour operators coordinate guides, transportation providers, attractions, and event organizers to deliver travel experiences. The company also supports customer loyalty programs, promotional campaigns, travel reviews, and personalized recommendations. Tourism managers need access to booking trends, customer preferences, seasonal demand patterns, and operational performance data across different destinations.

Group 10: National Sports League and Athlete Management System

A national sports authority oversees multiple professional leagues, tournaments, clubs, stadiums, training centers, media partners, sponsors, and sports academies. The system manages athletes, coaches, referees, teams, tournaments, matches, training programs, medical support services, sponsorship agreements, broadcasting rights, and fan engagement activities. Athletes participate in matches, training camps, fitness programs, talent development initiatives, and transfer processes between teams. Coaches develop strategies, conduct training sessions, evaluate player performance, and coordinate with medical staff. Tournament organizers schedule matches, assign officials, manage venues, and distribute awards and revenues. The authority also tracks injuries, disciplinary actions, player contracts, sponsorship deals, media coverage, ticket sales, and fan memberships. Sports administrators require extensive performance analytics and operational reports to support decision-making across the entire sporting ecosystem.